

Guideline Title: Acute Coronary Syndrome

Guideline Number: GUID-3203-C001

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
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- I. **PURPOSE:** To improve the health of the patient with myocardial ischemia and provide medical interventions to minimize the effects of a cardiac event. Acute coronary syndrome covers clinical conditions ranging from angina, unstable angina, non-Q-wave myocardial infarction, Q-wave myocardial infarction, and non-ST elevation myocardial infarction (NSTEMI). This guideline is for patients not going directly to the cath lab for immediate intervention.
- II. **DEPARTMENT GUIDELINE:**
 - a. Assess and manage airway, breathing and circulation.
 - b. Provide supplemental oxygen to maintain oxygen saturation > 93%.
 - c. Initiate cardiac monitoring, pulse ox, and serial vital signs. Apply End Tidal CO2 monitor to all intubated patients. Record readings.
 - d. Obtain/Evaluate 12 lead ECG and interpret findings. Assess for dysrhythmias and treat accordingly. Perform a repeat ECG if clinically indicated.
 - i. If STEMI indicated by ECG, refer to *GUID3203-C010 ST elevation myocardial infarction*
 - e. Ensure administration of **Aspirin 162-324 mg PO (chewable)** unless contraindicated by a true allergy or previously given within last 8 hours.
 - f. Establish IV access. (Consider 2 sites.)
 - g. If symptoms persist, consider administering **Nitroglycerine (NTG) 0.4 mg SL every 5 minutes x 3 doses** if hemodynamically stable. Use caution in inferior wall MI or right ventricular infarct in V4R.
 - i. If symptoms are relieved by SL NTG or if pain persists, consider initiating a **Nitroglycerin infusion at 10 mcg/min**, titrate until pain free, keeping MAP > 65 mmHg or SBP > 100 mmHg. Do not exceed 200 mcg/min.
 - ii. NTG is contraindicated if patient has received drugs for erectile dysfunction such as sildenafil (Viagra) or vardenafil (Levitra) within the past 24 hours and 72 hours for tadalafil (Cialis). (Consult medical direction).
 - h. If severe pain persists, consider administering:
 - i. **Morphine 0.1 mg/kg slow IV/IO** may repeat every 10 minutes PRN
 - ii. **Fentanyl 0.5-2 mcg/kg slow IV/IO** may repeat every 10 minutes PRN

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- iii. Monitor respiratory status and consider ETCO2 monitoring
- i. If the patient has not received low molecular weight heparin during the previous 12 hours and heparin therapy has not been initiated, consider administration of heparin after consulting with MD.
 - i. **Heparin bolus of 60 units/kg (max 5,000 units) followed by Heparin infusion at 18 units/kg (max 1,000 units per hour).**
- j. If patient's destination is not the cath lab, all IV medications such as low dose Nitroglycerin and Heparin should be continued.
- k. If glycoprotein IIB/IIIA inhibitor/Integrilin therapy has been initiated, continue infusion per referring physician's protocol.
- l. If hypotension is present, and there is no evidence of pulmonary edema, consider an IV fluid challenge of 250 ml 0.9% NS. Repeat as necessary.
- m. If hypotension persists, consider using vasopressors per *GUID3203-M011 Hypotension*.
- n. If patient is being transferred on an intra-aortic balloon pump refer to *GUID3203-PR023 Intra-Aortic Balloon Pump*.

III. DOCUMENTATION:

- a. EMS charts

IV. REFERENCES:

- a. American Heart Association, 2015.